RuralClass

Multi-grade classroom planning and student support

Foundry Games 2026 · Hackathon Level 1 · Category: Social · Educational Analytics / Recommendation AI

Problem Statement

The problem: In rural schools one teacher may handle several grades while planning lessons, tracking progress, and doing admin. There is no simple way to turn partial progress data into a realistic weekly plan.

Why it matters: Better planning and progress visibility make scarce teacher time count, and early detection helps before gaps grow.

Expected solution: A teacher-centered tool: enter grades, progress, lesson needs, and class time. AI suggests a weekly schedule, activities, resources, and intervention priorities; the teacher decides. The MVP can use a simulated classroom.

1. Problem Understanding

One teacher, many grades. In over a lakh Indian schools, a single teacher teaches several classes at once, while also planning lessons and doing admin work.

- Very little teaching time for each grade
- Lessons planned by guesswork
- Learning gaps found late. A Class 5 student weak in decimals may really be missing Class 3 fractions.
- Almost no multigrade training for teachers
- Most education apps are built for one-grade classes

2. Our Solution

What it is: RuralClass is a **weekly planning assistant for the teacher**. The teacher gives student marks and class time, and the app gives back a ready weekly plan. It does not teach students one by one. It plans the teacher's time.

How it works

1. **Teacher enters** student marks, pending topics, class time, and what the school has (chalk, paper, pebbles).

2. **App finds weak topics**, not weak students.
3. **App traces each gap to its root cause** (explained in the novelty below).
4. **App builds the weekly plan:** one shared lesson, a task for each grade, level groups, and peer pairs. The teacher picks from the suggested slots.
5. **AI adds low-cost activities** using what the school has. The teacher edits and approves.

What the teacher gets: a printable weekly timetable, a task for each grade, groups with peer pairs, and a short list of 3 or 4 students for quick check-ins.

Example. *Input:* Class 3 is weak in fractions, Class 4 in fractions and word problems, Class 5 in decimals. *Output:* one shared fractions lesson for Classes 3 to 5 at three depths, strong students paired with weaker ones, and 2 or 3 students flagged for a 5-minute check-in.

3. Novelty: Concept Bridge Map

Other tools plan one grade at a time. RuralClass uses what grades share. Topics build on each other across grades, for example Class 3 fractions → Class 4 fractions of a group → Class 5 decimals.

- **Root-cause detection:** finds *why* a grade is weak, not just where
- **One lesson, many grades:** one shared lesson replaces three, saving the teacher time
- **Built-in coach:** each plan says *why* it works, filling the missing multigrade training

This idea exists because the room is multigrade, so a normal single-grade planner cannot copy it.

4. Innovation and Creativity

- Plans around **topics and groups**, not individual students, which respects the teacher's time limit
- **Rules + AI:** simple rules analyse and schedule, and the LLM only writes activities and reasons
- **Low-resource activities** using only what the village school has
- **Printable plan.** No student devices needed.
- **Teacher stays in control.** The AI suggests, the teacher decides.

5. Social and Real-World Impact

- Supports **33 lakh+ children** in single-teacher schools
- Gives teachers back planning time

- Catches learning gaps early, before they grow across grades
- Helps the rural schools least likely to get new tools

6. Market Research

Fact	Source
1,04,125 single-teacher schools serve 33.8 lakh students (about 34 per school)	Ministry of Education, 2024-25
Over 100,000 single-teacher schools still exist	UDISE+ 2025-26
Multigrade rooms mean much less teaching time per grade	Ideas for India (small schools research)
No teacher education course in India focuses on multigrade teaching	Madhi Foundation

Gap in the market: AI planners and learning apps plan one grade at a time and often need student devices. Nobody plans *across* grades for the teacher.